13 - Final Year Viva Preparation

Questions grounded in this project

Knowledge-pack rule: every statement below is derived from files in this project or is explicitly labelled as planned, missing or not verified. Nothing here is an invented citation, dataset, result or feature.

Source files used for this document

all project files; 11_Monthly_Project_Reviews; 12_Experimental_Results_and_Conclusion

Answers refer to what the repository actually contains. Where you are unsure whether something has been run, check the verification table in document 01 first.

Questions and model answers

Q1. Why did you use synthetic data?

Real fraud and customer data are sensitive and not available for a student project; synthetic data lets experiments run in an isolated environment with known labels and a fixed seed. The limitation is that results validate the method, not real prevalence.

Q2. Why not use accuracy?

Fraud is about 2% of transactions, so predicting 'legitimate' every time scores ~98% accuracy while catching nothing. We use precision, recall, F1, PR-AUC and false-positive rate.

Q3. Why PR-AUC to pick the champion?

It focuses on the rare positive class, whereas ROC-AUC can look optimistic when negatives dominate (Saito and Rehmsmeier is cited in references.bib).

Q4. Why a chronological split?

Shuffled splits let the model learn from the future. Fraud evolves over time, so we train on the earliest 70%, tune the threshold on the next 15% and test once on the latest 15%.

Q5. How do you prevent leakage?

Customer-history features use cumulative sums minus the current row; a unit test changes the last transaction and checks earlier features do not change. fraud_pattern is never a feature.

Q6. Why class weights instead of SMOTE?

Resampling distorts predicted probabilities, which we use for risk tiers; class weights keep the data intact (Dal Pozzolo et al. discuss this in references.bib).

Q7. How is the threshold chosen?

Maximum F1 on the validation set, then fixed for the test set, so the test set is not used for tuning.

Q8. Explain Isolation Forest.

It builds random trees; anomalies are isolated in fewer splits. It needs no labels, which suits delayed-label situations; labels are used only afterwards to evaluate.

Q9. What is the false-positive rate and why does it matter for trust?

FP/(FP+TN): legitimate customers wrongly challenged. Each wrong block is friction that can reduce trust; the Platform Security Index penalises it.

Q10. What is the Consumer Trust Index?

A proposed composite: 0.6 x mean survey trust score + 0.4 x Platform Security Index. Weights are design choices; the project reports sensitivity but has not validated the index psychometrically.

Q11. Is your survey real?

No. The bundled survey_data.csv is synthetic. The questionnaire and ethics checklist are real design artefacts; real collection needs ethics approval, which is not recorded.

Q12. What does Cronbach's alpha tell you?

Internal consistency of the items (Cronbach 1951); about 0.7 or higher is conventionally acceptable. On synthetic data it reflects the generator, not real respondents.

Q13. How does Grafana show real data?

The worker computes metrics from the pipeline and from a replay of held-out transactions through the trained model, and Prometheus scrapes them. Panels are PromQL, LogQL or SQL queries, not typed values.

Q14. What is the live replay?

After training, the worker feeds held-out test transactions in batches through the champion model and increments TP/FP/FN/TN counters. It is a replay, not production traffic.

Q15. Why Prometheus, Loki and PostgreSQL together?

Prometheus for numeric time series, Loki for logs, PostgreSQL for structured results and run history; Grafana can query all three.

Q16. What happens on docker compose up?

postgres starts and applies the schema; the worker builds data, trains, evaluates, writes artefacts and publishes metrics, becoming healthy when fintech_pipeline_ready is 1; then jupyter starts. Prometheus, Loki, Promtail, cAdvisor and Grafana start with health checks.

Q17. How is reproducibility ensured?

Seeds in settings.yaml, data hash and library versions saved in model metadata, containers, tests. With end_date 'auto' the dates move with the run date but values are the same.

Q18. Which model won?

In the last run the champion was xgboost. State the numbers from document 12 and mention the small number of test frauds and single seed.

Q19. What are the main limitations?

Synthetic data and survey, single seed, small tuning budget, no drift, unvalidated index, no ethics approval recorded, replay not production.

Q20. What would you do next?

Run the ULB dataset, repeat over seeds, collect real consented survey data, add calibration and SHAP, drift monitoring and Alertmanager.

Q21. How did the project progress over four reviews?

Review 1: problem, objectives, literature, gap, plan. Review 2: methodology, architecture, data, preprocessing and foundations. Review 3: models, trust scoring, dashboards and preliminary results. Review 4: final comparison, dashboards, conclusions.

Honesty checklist before the viva

- Say clearly that data and survey are synthetic.
- Do not quote results without the seed, test-set size and uncertainty.
- Do not claim ethics approval or real respondents.
- Be ready to show: tests passing, pipeline summary, one Grafana dashboard live.
- Know what is planned versus implemented (documents 01, 07, 08, 10).

Demo plan

- docker compose up --build; wait for worker healthy.
- Grafana Overview -> Pipeline Monitoring -> Fraud -> Model Performance -> Trust -> Infrastructure.
- Jupyter notebook 07 for evaluation; open a Loki log line.
- Fallback if the stack is slow: show notebooks and report/figures.